

A Level Mathematics 9709 · P3 Pure Mathematics 3

全套中英文详细学习手册

Detailed bilingual student manual · concept, derivation, worked examples
and exam method

本册共 10 个独立课时。每课按三页展开：第一页建立概念、推导方法并完成基础例题；第二页处理 Cambridge 风格应用、英文表达、评分点与易错辨析；第三页扩展考纲边界、变式推演与迁移任务。题目均为依据考纲原创的训练示例，不复制历年试题。

使用范围与版本 | Scope

0580 册依据 Cambridge IGCSE Mathematics 0580 (2025/2027) 的 Core/Extended 分层；9709 册依据 Cambridge International AS & A Level Mathematics 9709 (2026/2027)。M2 单独标为拓展/旧体系衔接材料，不作为现行独立考试组件。

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2. 模、指数与对数深化 | Modulus, exponential and logarithmic functions
3. 三角函数深化 | Advanced trigonometry
4. 复杂微分 | Further differentiation
5. 复杂积分 | Further integration
6. 微分方程 | Differential equations
7. 复数 | Complex numbers
8. 向量三维 | Vectors in 3D
9. 数值积分 | Numerical integration
10. P3 综合策略 | P3 synthesis

第 1 课 | 多项式与部分分式 | Polynomials and partial fractions

Lesson 1 of 10 · Detailed bilingual learning unit

概念拆解 | Concept and meaning

因式定理、余式定理与多项式除法用于拆解高次式；部分分式把有理式化为可积分形式。

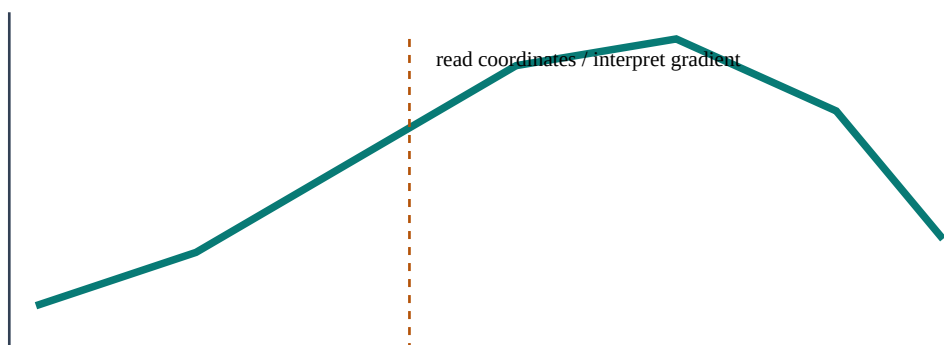
English core explanation. Algebra records a sequence of equivalent statements. Show the operation that preserves equality, distinguish an expression from an equation, and check restrictions caused by denominators, logarithms, roots or modulus. Substitute solutions back into the original statement.

为什么方法成立 | Reasoning behind the method

先确认真分式；分母不同因子决定分子形式，重复因子需逐级列项。

把知识点拆成可执行步骤 | A reliable method

1. Read the command word and identify the target quantity. 先圈出 find, show, prove, sketch, hence 或 explain。
2. Translate the information into equations, a labelled diagram, a table or a distribution.
3. Choose a theorem or formula only after checking its assumptions.
4. Keep exact values through intermediate steps; round once at the end.
5. Check sign, size, unit, domain and whether all solutions have been found.



例题一：从题意到完整解答 | Worked example 1

分解 $(5x+1)/[(x+1)(x+2)]$ 。

令 $A/(x+1)+B/(x+2)$ ，得 $5x+1=A(x+2)+B(x+1)$ 。代 $x=2$ 得 $B=11/3$ ； $x=1$ 得 $A=4/3$ 。

Reading the mathematics. The first line identifies the structure; the middle lines show the transformation or calculation; the final line answers the question in context. Do not hide essential reasoning inside the calculator.

第 1 课（续） | 多项式与部分分式 | Polynomials and partial fractions

例题二：Cambridge 风格应用 | Cambridge-style application

Find remainder when $p(x)=x^2x+5$ is divided by x^2 .

Remainder= $p(2)=84+5=9$.

评分点如何产生 | Where marks come from

多数结构题把分数分给不同层次：建立正确模型或公式通常得到 method mark；正确代入并推进得到后续 method/accuracy marks；准确答案、单位、范围或结论得到 final accuracy

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follow-through，因此必须写出过程。证明题不能只写计算器结果，show that 题还要到达题目指定的精确形式。

English mathematical communication

Use complete mathematical sentences: “Substituting ... gives ...”, “Therefore ...”, “Since the discriminant is zero ...”, “The events are independent because ...”, or “The negative sign shows motion opposite to the chosen positive direction.” When a question says hence, use the previous result. When it says exact, do not replace π , e , logarithms or radicals by rounded decimals.

易错辨析 | Common errors and repairs

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Exam-ready summary. Define or identify $\rightarrow$ model $\rightarrow$ calculate with visible steps $\rightarrow$ interpret $\rightarrow$ check. Your final statement should answer the exact question asked, with appropriate accuracy and units.

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Detailed English explanation. Algebra records a sequence of equivalent statements. Show the operation that preserves equality, distinguish an expression from an equation, and check restrictions caused by denominators, logarithms, roots or modulus. Substitute solutions back into the original statement.

变式推演 | Variation and transfer

若把例题中的具体数换成字母，应先建立一般关系再代值；若改变范围、方向、图形性质或随机条件，则必须重新检查原方法的适用条件。把答案遮住，尝试只凭题干写出第一行模型；第一行正确，后续计算才有方向。

第三道示范：审题、计算、复核 | Worked reasoning example

重新审视本课基础题：分解 $(5x+1)/[(x+1)(x+2)]$ 。

先把文字条件转换成数学关系，再依据本课方法推进。完整结果是：令 $\frac{5x+1}{(x+1)(x+2)} = \frac{A}{x+1} + \frac{B}{x+2}$ ，得 $5x+1 = A(x+2) + B(x+1)$ 。代 $x=2$ 得 $B=11/3$ ； $x=1$ 得 $A=4/3$ 。复核时从结论逆向代回，并检查量级、正负、单位和定义域。若不一致，优先检查括号、指数、角度模式、方向约定或概率事件范围。

Cambridge 题型如何包装 | How the idea is examined

选择题常把典型失误设计成干扰项：漏掉第二个解、混淆百分数基数、把面积比当长度比、把独立当互斥或遗漏连续性修正。结构题常先要求 show/find 一个中间结果，再用 hence 推进。上一问的精确形式通常是下一问最简洁的入口。

英文作答骨架 | Useful response frames

“From the given information, ...” → “Using ..., we obtain ...” → “Substituting and simplifying gives ...”
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第三个迁移任务 | Transfer task

Find remainder when $p(x)=x^2x+5$ is divided by x^2 .

独立完成后对照：Remainder= $p(2)=84+5=9$.

Exam check. Have you answered the command word, shown the decisive line of working, kept suitable accuracy, included units or direction, and considered every valid solution?

第 2 课 | 模、指数与对数深化 | Modulus, exponential and logarithmic functions

Lesson 2 of 10 · Detailed bilingual learning unit

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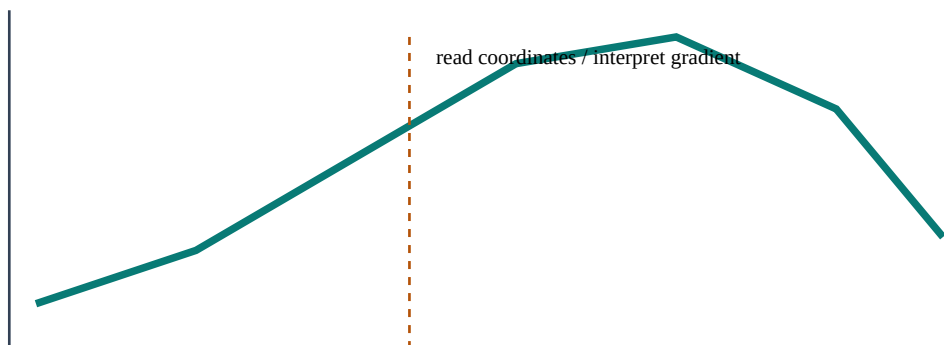
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为什么方法成立 | Reasoning behind the method

解方程要结合图像交点和定义域；换底处理不同底数。

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解 $e^{(2x)}=7$ 。

$2x=\ln 7$, $x=0.5\ln 7$ 。

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第 2 课（续） | 模、指数与对数深化 | Modulus, exponential and logarithmic functions

例题二：Cambridge 风格应用 | Cambridge-style application

Solve $|x+4|=3$.

$x+4=3$ gives $x=\pm \sqrt{7}$; $x+4=3$ gives $x=\pm 1$.

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Solve $|x^4|=3$.

独立完成后对照： $x^4=3$ gives $x=\pm \sqrt[4]{3}$; $x^4=3$ gives $x=\pm 1$.

Exam check. Have you answered the command word, shown the decisive line of working, kept suitable accuracy, included units or direction, and considered every valid solution?

第 3 课 | 三角函数深化 | Advanced trigonometry

Lesson 3 of 10 · Detailed bilingual learning unit

概念拆解 | Concept and meaning

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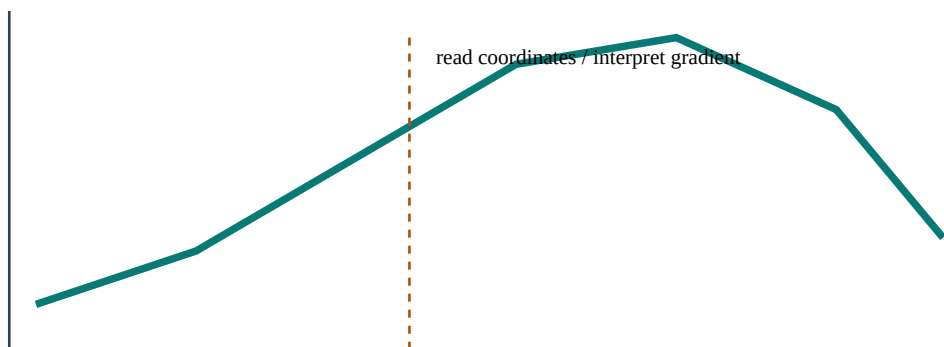
English core explanation. Use more than one representation. A formula gives exact relationships, a graph shows shape and intersections, and a table reveals numerical patterns. Factorised, expanded and completed-square forms each expose different information.

为什么方法成立 | Reasoning behind the method

$R = \sqrt{a^2 + b^2}$, 比较系数求 α , 随后处理值域。

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例题一：从题意到完整解答 | Worked example 1

写 $3\sin x + 4\cos x$ 为 $R\sin(x + \alpha)$ 。

$R=5$, $5\cos \alpha=3$, $5\sin \alpha=4$, 所以 $\alpha=\tan^{-1}(4/3)$ 。

Reading the mathematics. The first line identifies the structure; the middle lines show the transformation or calculation; the final line answers the question in context. Do not hide essential reasoning inside the calculator.

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Hence state the maximum value.

Since $\sin(x + \alpha) \leq 1$, the maximum is 5.

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第 4 课 | 复杂微分 | Further differentiation

Lesson 4 of 10 · Detailed bilingual learning unit

概念拆解 | Concept and meaning

隐函数、参数方程和二阶导数扩展变化率分析。

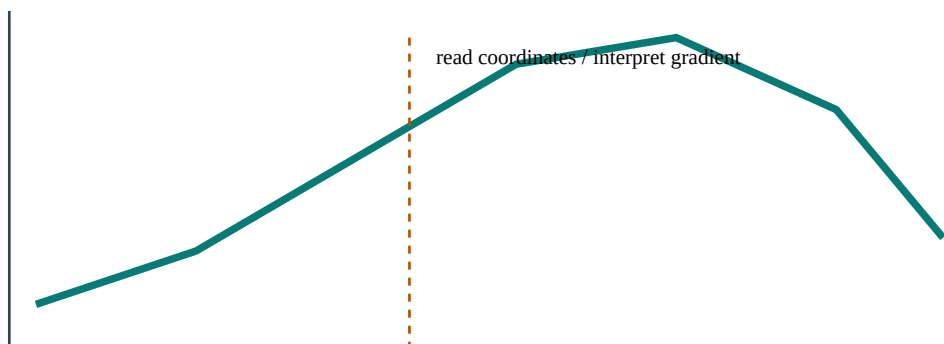
English core explanation. Calculus links local change with accumulated change. Identify the inner function or variable, retain the constant of integration when required, and interpret derivatives, stationary points and definite integrals rather than treating them as symbol manipulation.

为什么方法成立 | Reasoning behind the method

隐式微分每个y项乘 dy/dx ；参数式 $dy/dx=(dy/dt)/(dx/dt)$ 。

把知识点拆成可执行步骤 | A reliable method

1. Read the command word and identify the target quantity. 先圈出 find, show, prove, sketch, hence 或 explain。
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例题一：从题意到完整解答 | Worked example 1

$x+y=25$, 求 dy/dx 。

$2x+2y \ dy/dx=0$, 所以 $dy/dx=x/y$ 。

Reading the mathematics. The first line identifies the structure; the middle lines show the transformation or calculation; the final line answers the question in context. Do not hide essential reasoning inside the calculator.

第 4 课（续） | 复杂微分 | Further differentiation

例题二：Cambridge 风格应用 | Cambridge-style application

Given $x=t+1, y=t$, find dy/dx .

$dy/dx=(3t)/(2t)=3t/2$, for $t \neq 0$.

评分点如何产生 | Where marks come from

多数结构题把分数分给不同层次：建立正确模型或公式通常得到 method mark；正确代入并推进得到后续 method/accuracy marks；准确答案、单位、范围或结论得到 final accuracy mark。若前一步算错但方法仍一致，后续常可获得 follow-through，因此必须写出过程。证明题不能只写计算器结果，show that 题还要到达题目指定的精确形式。

English mathematical communication

Use complete mathematical sentences: “Substituting ... gives ...”, “Therefore ...”, “Since the discriminant is zero ...”, “The events are independent because ...”, or “The negative sign shows motion opposite to the chosen positive direction.” When a question says hence, use the previous result. When it says exact, do not replace π , e , logarithms or radicals by rounded decimals.

易错辨析 | Common errors and repairs

条件未检查：把只适用于直角三角形、恒加速度、独立事件或正态模型的公式随意套用。修正：公式前写出适用结构。

中途过早取整：后续答案超出允许误差。修正：保留精确值或计算器完整储存值。

只给答案：丢失方法分。修正：至少展示关键公式、代入与一次等价变形。

符号或单位遗漏：尤其是向量方向、平方/立方单位、概率区间和角度模式。

漏解或增根：解三角、模、平方与对数方程后，回到原定义域逐一检查。

再推一步 | Extension and connection

把本课与相邻知识连接：同一对象通常可从符号、图像、数值表和现实情境四种表示来理解。若代数式得到的符号与图像趋势不一致，应回查运算；若概率或测量答案超出合理范围，应先检查模型而不是盲信计算器。这样的交叉检查是从“会算”走向“会解题”的关键。

Exam-ready summary. Define or identify $\rightarrow$ model $\rightarrow$ calculate with visible steps $\rightarrow$ interpret $\rightarrow$ check. Your final statement should answer the exact question asked, with appropriate accuracy and units.

知识边界与相邻考点 | Scope and connections

隐函数、参数方程和二阶导数扩展变化率分析。考试不会总以熟悉的标题出现：它可能被包装成图像、表格、坐标、单位、参数或现实情境。先还原“已知量—未知量—约束关系”，再决定使用哪条规则。隐式微分每个 y 项乘 dy/dx ；参数式 $dy/dx=(dy/dt)/(dx/dt)$ 。

Detailed English explanation. Calculus links local change with accumulated change. Identify the inner function or variable, retain the constant of integration when required, and interpret derivatives, stationary points and definite integrals rather than treating them as symbol manipulation.

变式推演 | Variation and transfer

若把例题中的具体数换成字母，应先建立一般关系再代值；若改变范围、方向、图形性质或随机条件，则必须重新检查原方法的适用条件。把答案遮住，尝试只凭题干写出第一行模型；第一行正确，后续计算才有方向。

第三道示范：审题、计算、复核 | Worked reasoning example

重新审视本课基础题： $x+y=25$ ，求 dy/dx 。

先把文字条件转换成数学关系，再依据本课方法推进。完整结果是： $2x+2y \, dy/dx=0$ ，所以 $dy/dx=-x/y$ 。复核时从结论逆向代回，并检查量级、正负、单位和定义域。若不一致，优先检查括号、指数、角度模式、方向约定或概率事件范围。

Cambridge 题型如何包装 | How the idea is examined

选择题常把典型失误设计成干扰项：漏掉第二个解、混淆百分数基数、把面积比当长度比、把独立当互斥或遗漏连续性修正。结构题常先要求 show/find 一个中间结果，再用 hence 推进。上一问的精确形式通常是下一问最简洁的入口。

英文作答骨架 | Useful response frames

“From the given information, ...” → “Using ..., we obtain ...” → “Substituting and simplifying gives ...” → “Therefore the required value is ...”. For interpretation, write “This means that ... in the context of the question.” For validation, write “The value lies in the stated domain and satisfies the original condition.”

第三个迁移任务 | Transfer task

Given $x=t+1, y=t$, find dy/dx .

独立完成后对照： $dy/dx=(3t)/(2t)=3t/2$, for $t \neq 0$.

Exam check. Have you answered the command word, shown the decisive line of working, kept suitable accuracy, included units or direction, and considered every valid solution?

第 5 课 | 复杂积分 | Further integration

Lesson 5 of 10 · Detailed bilingual learning unit

概念拆解 | Concept and meaning

换元、分部积分和部分分式处理非直接积分。

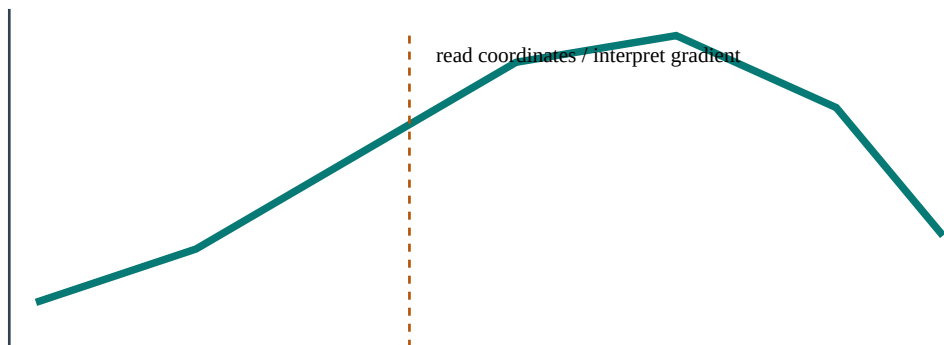
English core explanation. Calculus links local change with accumulated change. Identify the inner function or variable, retain the constant of integration when required, and interpret derivatives, stationary points and definite integrals rather than treating them as symbol manipulation.

为什么方法成立 | Reasoning behind the method

看到复合及其导数考虑换元；多项式 × 指数/三角常用分部积分。

把知识点拆成可执行步骤 | A reliable method

1. Read the command word and identify the target quantity. 先圈出 find, show, prove, sketch, hence 或 explain。
2. Translate the information into equations, a labelled diagram, a table or a distribution.
3. Choose a theorem or formula only after checking its assumptions.
4. Keep exact values through intermediate steps; round once at the end.
5. Check sign, size, unit, domain and whether all solutions have been found.



例题一：从题意到完整解答 | Worked example 1

计算 $\int x e^x dx$ 。

取 $u=x, dv=e^x dx$, 得 $x e^x - \int e^x dx = e^x(x-1) + C$ 。

Reading the mathematics. The first line identifies the structure; the middle lines show the transformation or calculation; the final line answers the question in context. Do not hide essential reasoning inside the calculator.

第 5 课（续） | 复杂积分 | Further integration

例题二：Cambridge 风格应用 | Cambridge-style application

Integrate $2x/(x+3)$.

Let $u=x+3$, $du=dx$; $\text{integral}=\ln(x+3)+C$.

评分点如何产生 | Where marks come from

多数结构题把分数分给不同层次：建立正确模型或公式通常得到 method mark；正确代入并推进得到后续 method/accuracy marks；准确答案、单位、范围或结论得到 final accuracy mark。若前一步算错但方法仍一致，后续常可获得 follow-through，因此必须写出过程。证明题不能只写计算器结果，show that 题还要到达题目指定的精确形式。

English mathematical communication

Use complete mathematical sentences: “Substituting ... gives ...”, “Therefore ...”, “Since the discriminant is zero ...”, “The events are independent because ...”, or “The negative sign shows motion opposite to the chosen positive direction.” When a question says hence, use the previous result. When it says exact, do not replace π , e , logarithms or radicals by rounded decimals.

易错辨析 | Common errors and repairs

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换元、分部积分和部分分式处理非直接积分。考试不会总以熟悉的标题出现：它可能被包装成图像、表格、坐标、单位、参数或现实情境。先还原“已知量—未知量—约束关系”，再决定使用哪条规则。看到复合及其导数考虑换元；多项式 $\times$ 指数/三角常用分部积分。

Detailed English explanation. Calculus links local change with accumulated change. Identify the inner function or variable, retain the constant of integration when required, and interpret derivatives, stationary points and definite integrals rather than treating them as symbol manipulation.

变式推演 | Variation and transfer

若把例题中的具体数换成字母，应先建立一般关系再代值；若改变范围、方向、图形性质或随机条件，则必须重新检查原方法的适用条件。把答案遮住，尝试只凭题干写出第一行模型；第一行正确，后续计算才有方向。

第三道示范：审题、计算、复核 | Worked reasoning example

重新审视本课基础题：计算 $\int x e^x dx$ 。

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Cambridge 题型如何包装 | How the idea is examined

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第三个迁移任务 | Transfer task

Integrate $2x/(x+3)$.

独立完成后对照：Let $u=x+3, du=dx$; $\int \frac{2x}{x+3} dx = 2 \int \frac{x+3-3}{x+3} dx = 2 \int \frac{x+3}{x+3} dx - 6 \int \frac{1}{x+3} dx = 2x - 6 \ln|x+3| + C$.

Exam check. Have you answered the command word, shown the decisive line of working, kept suitable accuracy, included units or direction, and considered every valid solution?

第 6 课 | 微分方程 | Differential equations

Lesson 6 of 10 · Detailed bilingual learning unit

概念拆解 | Concept and meaning

一阶可分离微分方程描述变化率随变量变化；积分常数由初始条件确定。

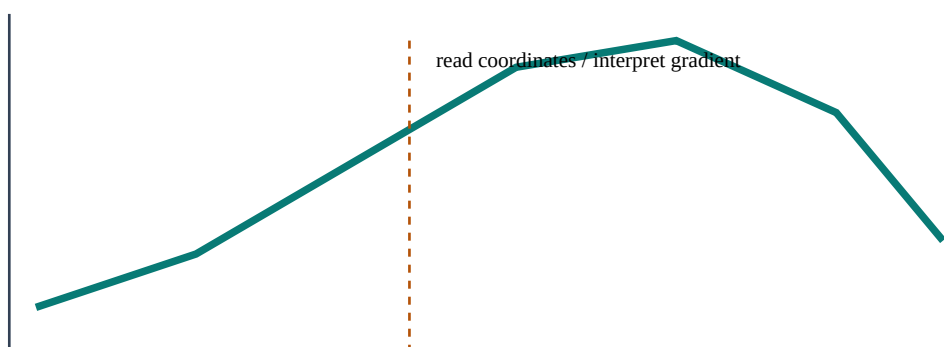
English core explanation. Algebra records a sequence of equivalent statements. Show the operation that preserves equality, distinguish an expression from an equation, and check restrictions caused by denominators, logarithms, roots or modulus. Substitute solutions back into the original statement.

为什么方法成立 | Reasoning behind the method

把y项与x项分到两边，分别积分，再代初值。

把知识点拆成可执行步骤 | A reliable method

1. Read the command word and identify the target quantity. 先圈出 find, show, prove, sketch, hence 或 explain。
2. Translate the information into equations, a labelled diagram, a table or a distribution.
3. Choose a theorem or formula only after checking its assumptions.
4. Keep exact values through intermediate steps; round once at the end.
5. Check sign, size, unit, domain and whether all solutions have been found.



例题一：从题意到完整解答 | Worked example 1

$\frac{dy}{dx}=3xy$, $y=2$ 当 $x=0$, 求 y 。

$\frac{dy}{y}=3x \, dx$; $\ln y=1.5x+C$; $C=\ln 2$, 所以 $y=2e^{(1.5x)}$ 。

Reading the mathematics. The first line identifies the structure; the middle lines show the transformation or calculation; the final line answers the question in context. Do not hide essential reasoning inside the calculator.

第 6 课（续） | 微分方程 | Differential equations

例题二：Cambridge 风格应用 | Cambridge-style application

Why is an arbitrary constant necessary?

Differentiation removes constants; integration represents a family of solutions until an initial condition selects one.

评分点如何产生 | Where marks come from

多数结构题把分数分给不同层次：建立正确模型或公式通常得到 method mark；正确代入并推进得到后续 method/accuracy marks；准确答案、单位、范围或结论得到 final accuracy mark。若前一步算错但方法仍一致，后续常可获得

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English mathematical communication

Use complete mathematical sentences: “Substituting ... gives ...”, “Therefore ...”, “Since the discriminant is zero ...”, “The events are independent because ...”, or “The negative sign shows motion opposite to the chosen positive direction.” When a question says hence, use the previous result. When it says exact, do not replace π , e , logarithms or radicals by rounded decimals.

易错辨析 | Common errors and repairs

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Exam-ready summary. Define or identify $\rightarrow$ model $\rightarrow$ calculate with visible steps $\rightarrow$ interpret $\rightarrow$ check. Your final statement should answer the exact question asked, with appropriate accuracy and units.

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Detailed English explanation. Algebra records a sequence of equivalent statements. Show the operation that preserves equality, distinguish an expression from an equation, and check restrictions caused by denominators, logarithms, roots or modulus. Substitute solutions back into the original statement.

变式推演 | Variation and transfer

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第三道示范：审题、计算、复核 | Worked reasoning example

重新审视本课基础题： $dy/dx=3xy$ ， $y=2$ 当 $x=0$ ，求 y 。

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 dx ； $\ln y=1.5x+C$ ； $C=\ln 2$ ，所以 $y=2e^{(1.5x)}$ 。复核时从结论逆向代回，并检查量级、正负、单位和定义域。若不一致，优先检查括号、指数、角度模式、方向约定或概率事件范围。

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第三个迁移任务 | Transfer task

Why is an arbitrary constant necessary?

独立完成后对照：Differentiation removes constants; integration represents a family of solutions until an initial condition selects one.

Exam check. Have you answered the command word, shown the decisive line of working, kept suitable accuracy, included units or direction, and considered every valid solution?

第 7 课 | 复数 | Complex numbers

Lesson 7 of 10 · Detailed bilingual learning unit

概念拆解 | Concept and meaning

$i=1$; Argand 图把 $z=x+iy$ 视为点; 模与辐角给极形式。

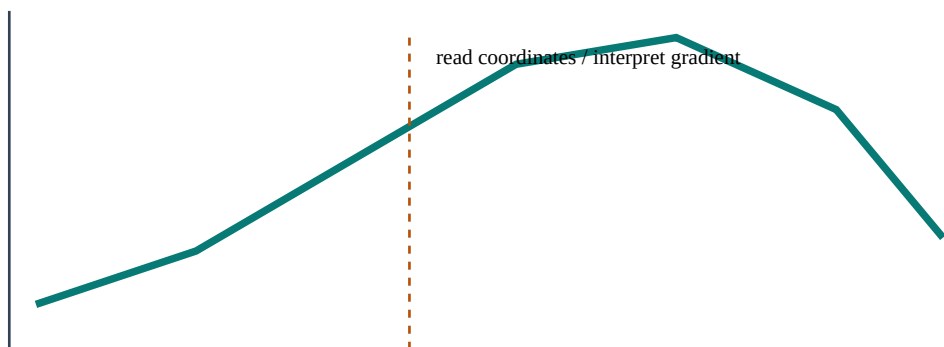
English core explanation. Direction and order matter. A transformation must be described with every defining parameter; vector subtraction gives directed displacement; matrix multiplication acts in a fixed order; complex modulus and argument encode size and direction.

为什么方法成立 | Reasoning behind the method

共轭用于实化分母; 乘法在极形式中模相乘、辐角相加。

把知识点拆成可执行步骤 | A reliable method

1. Read the command word and identify the target quantity. 先圈出 find, show, prove, sketch, hence 或 explain。
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例题一: 从题意到完整解答 | Worked example 1

化简 $(3+2i)/(1i)$ 。

乘 $(1+i)$: $[(3+2i)(1+i)]/2=(1+5i)/2$ 。

Reading the mathematics. The first line identifies the structure; the middle lines show the transformation or calculation; the final line answers the question in context. Do not hide essential reasoning inside the calculator.

第 7 课（续） | 复数 | Complex numbers

例题二：Cambridge 风格应用 | Cambridge-style application

Find modulus and argument of $1+i$.

$|z|=\sqrt{2}$ and $\arg z=\pi/4$.

评分点如何产生 | Where marks come from

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若把例题中的具体数换成字母, 应先建立一般关系再代值; 若改变范围、方向、图形性质或随机条件, 则必须重新检查原方法的适用条件。把答案遮住, 尝试只凭题干写出第一行模型; 第一行正确, 后续计算才有方向。

第三道示范: 审题、计算、复核 | Worked reasoning example

重新审视本课基础题: 化简 $(3+2i)/(1i)$ 。

先把文字条件转换成数学关系, 再依据本课方法推进。完整结果是: 乘 $(1+i)$: $[(3+2i)(1+i)]/2=(1+5i)/2$ 。复核时从结论逆向代回, 并检查量级、正负、单位和定义域。若不一致, 优先检查括号、指数、角度模式、方向约定或概率事件范围。

Cambridge 题型如何包装 | How the idea is examined

选择题常把典型失误设计成干扰项: 漏掉第二个解、混淆百分数基数、把面积比当长度比、把独立当互斥或遗漏连续性修正。结构题常先要求 show/find 一个中间结果, 再用 hence 推进。上一问的精确形式通常是下一问最简洁的入口。

英文作答骨架 | Useful response frames

“From the given information, ...” → “Using ..., we obtain ...” → “Substituting and simplifying gives ...”
→ “Therefore the required value is ...”. For interpretation, write “This means that ... in the context of the question.” For validation, write “The value lies in the stated domain and satisfies the original condition.”

第三个迁移任务 | Transfer task

Find modulus and argument of $1+i$.

独立完成后对照: $|z|=\sqrt{2}$ and $\arg z=\pi/4$.

Exam check. Have you answered the command word, shown the decisive line of working, kept suitable accuracy, included units or direction, and considered every valid solution?

第 8 课 | 向量三维 | Vectors in 3D

Lesson 8 of 10 · Detailed bilingual learning unit

概念拆解 | Concept and meaning

直线向量方程 $r=a+\lambda b$; 交点要求同一点同时满足两条线。

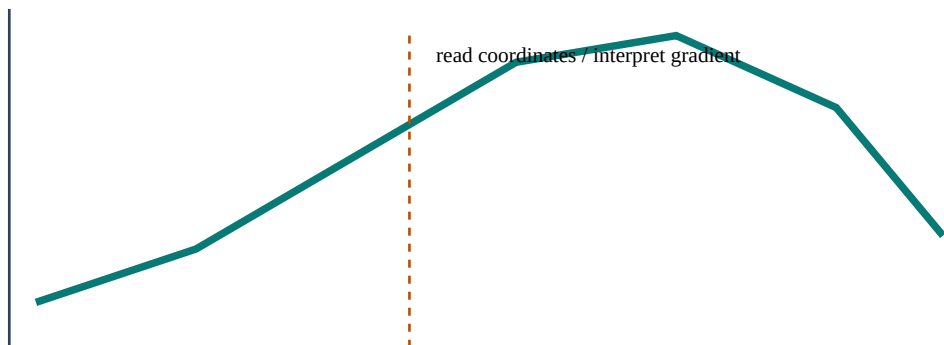
English core explanation. Direction and order matter. A transformation must be described with every defining parameter; vector subtraction gives directed displacement; matrix multiplication acts in a fixed order; complex modulus and argument encode size and direction.

为什么方法成立 | Reasoning behind the method

平行看方向向量成比例; 垂直用点积为0。

把知识点拆成可执行步骤 | A reliable method

1. Read the command word and identify the target quantity. 先圈出 find, show, prove, sketch, hence 或 explain。
2. Translate the information into equations, a labelled diagram, a table or a distribution.
3. Choose a theorem or formula only after checking its assumptions.
4. Keep exact values through intermediate steps; round once at the end.
5. Check sign, size, unit, domain and whether all solutions have been found.



例题一：从题意到完整解答 | Worked example 1

求 (1,2,3) 与 (4,1,5) 距离。

位移 (3,3,2), 距离 $\sqrt{9+9+4}=\sqrt{22}$ 。

Reading the mathematics. The first line identifies the structure; the middle lines show the transformation or calculation; the final line answers the question in context. Do not hide essential reasoning inside the calculator.

第 8 课（续） | 向量三维 | Vectors in 3D

例题二：Cambridge 风格应用 | Cambridge-style application

Show vectors $(1,2,1)$ and $(2,1,0)$ are perpendicular.

Dot product $= 2 + 0 = 0$.

评分点如何产生 | Where marks come from

多数结构题把分数分给不同层次：建立正确模型或公式通常得到 method mark；正确代入并推进得到后续 method/accuracy marks；准确答案、单位、范围或结论得到 final accuracy

mark。若前一步算错但方法仍一致，后续常可获得

follow-through，因此必须写出过程。证明题不能只写计算器结果，show that 题还要到达题目指定的精确形式。

English mathematical communication

Use complete mathematical sentences: “Substituting ... gives ...”, “Therefore ...”, “Since the discriminant is zero ...”, “The events are independent because ...”, or “The negative sign shows motion opposite to the chosen positive direction.” When a question says hence, use the previous result. When it says exact, do not replace π , e , logarithms or radicals by rounded decimals.

易错辨析 | Common errors and repairs

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中途过早取整：后续答案超出允许误差。修正：保留精确值或计算器完整储存值。

只给答案：丢失方法分。修正：至少展示关键公式、代入与一次等价变形。

符号或单位遗漏：尤其是向量方向、平方/立方单位、概率区间和角度模式。

漏解或增根：解三角、模、平方与对数方程后，回到原定义域逐一检查。

再推一步 | Extension and connection

把本课与相邻知识连接：同一对象通常可从符号、图像、数值表和现实情境四种表示来理解。若代数式得到的符号与图像趋势不一致，应回查运算；若概率或测量答案超出合理范围，应先检查模型而不是盲信计算器。这样的交叉检查是从“会算”走向“会解题”的关键。

Exam-ready summary. Define or identify $\rightarrow$ model $\rightarrow$ calculate with visible steps $\rightarrow$ interpret $\rightarrow$ check. Your final statement should answer the exact question asked, with appropriate accuracy and units.

知识边界与相邻考点 | Scope and connections

直线向量方程 $r=a+\lambda b$ ；交点要求同一点同时满足两条线。考试不会总以熟悉的标题出现：它可能被包装成图像、表格、坐标、单位、参数或现实情境。先还原“已知量—未知量—约束关系”，再决定使用哪条规则。平行看方向向量成比例；垂直用点积为0。

Detailed English explanation. Direction and order matter. A transformation must be described with every defining parameter; vector subtraction gives directed displacement; matrix multiplication acts in a fixed order; complex modulus and argument encode size and direction.

变式推演 | Variation and transfer

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第三道示范：审题、计算、复核 | Worked reasoning example

重新审视本课基础题：求(1,2,3)与(4,1,5)距离。

先把文字条件转换成数学关系，再依据本课方法推进。完整结果是：位移(3,3,2)，距离 $\sqrt{(9+9+4)}=\sqrt{22}$ 。复核时从结论逆向代回，并检查量级、正负、单位和定义域。若不一致，优先检查括号、指数、角度模式、方向约定或概率事件范围。

Cambridge 题型如何包装 | How the idea is examined

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第三个迁移任务 | Transfer task

Show vectors (1,2,1) and (2,1,0) are perpendicular.

独立完成后对照：Dot product=22+0=0.

Exam check. Have you answered the command word, shown the decisive line of working, kept suitable accuracy, included units or direction, and considered every valid solution?

第 9 课 | 数值积分 | Numerical integration

Lesson 9 of 10 · Detailed bilingual learning unit

概念拆解 | Concept and meaning

梯形法用直线段近似曲线下面积，分割越细通常越准确。

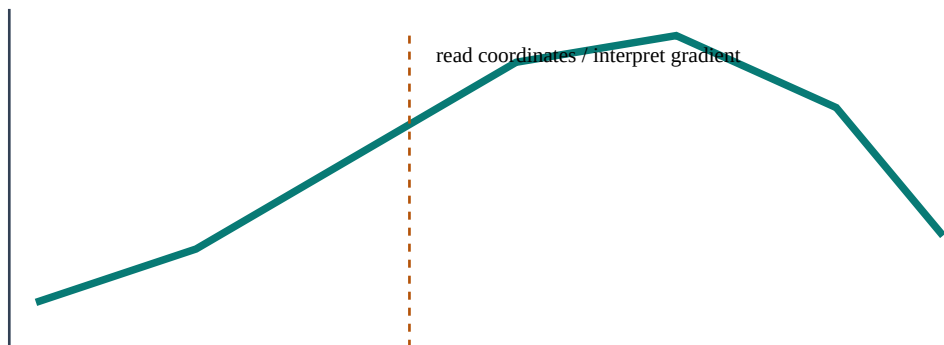
English core explanation. Calculus links local change with accumulated change. Identify the inner function or variable, retain the constant of integration when required, and interpret derivatives, stationary points and definite integrals rather than treating them as symbol manipulation.

为什么方法成立 | Reasoning behind the method

$h=(b-a)/n$; 端点一次、中间点两次。

把知识点拆成可执行步骤 | A reliable method

1. Read the command word and identify the target quantity. 先圈出 find, show, prove, sketch, hence 或 explain。
2. Translate the information into equations, a labelled diagram, a table or a distribution.
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4. Keep exact values through intermediate steps; round once at the end.
5. Check sign, size, unit, domain and whether all solutions have been found.



例题一：从题意到完整解答 | Worked example 1

用 $n=2$ 梯形估算 $\int x dx$ 。

$h=1$, 估算 $=1/2[0+2(1)+4]=3$ 。

Reading the mathematics. The first line identifies the structure; the middle lines show the transformation or calculation; the final line answers the question in context. Do not hide essential reasoning inside the calculator.

第 9 课（续） | 数值积分 | Numerical integration

例题二：Cambridge 风格应用 | Cambridge-style application

State why this is an overestimate.

For a convex curve $y=x$, the chords lie above the curve.

评分点如何产生 | Where marks come from

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English mathematical communication

Use complete mathematical sentences: “Substituting ... gives ...”, “Therefore ...”, “Since the discriminant is zero ...”, “The events are independent because ...”, or “The negative sign shows motion opposite to the chosen positive direction.” When a question says hence, use the previous result. When it says exact, do not replace π , e , logarithms or radicals by rounded decimals.

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第三道示范：审题、计算、复核 | Worked reasoning example

重新审视本课基础题：用 $n=2$ 梯形估算 $\int x dx$ 。

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Cambridge 题型如何包装 | How the idea is examined

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第三个迁移任务 | Transfer task

State why this is an overestimate.

独立完成后对照：For a convex curve $y=x$, the chords lie above the curve.

Exam check. Have you answered the command word, shown the decisive line of working, kept suitable accuracy, included units or direction, and considered every valid solution?

第 10 课 | P3 综合策略 | P3 synthesis

Lesson 10 of 10 · Detailed bilingual learning unit

概念拆解 | Concept and meaning

P3 题常让前一小问提供后一问可用的恒等式或分解。

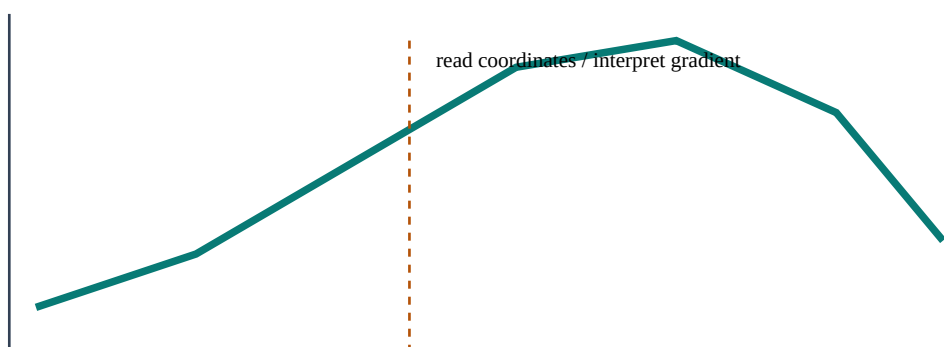
English core explanation. Translate between words, symbols, diagrams and numerical values. State assumptions, keep exact values during working, and test the final result against the original domain, scale and context.

为什么方法成立 | Reasoning behind the method

先读完整题，保留题设形式；每一步检查定义域、参数与精度。

把知识点拆成可执行步骤 | A reliable method

1. Read the command word and identify the target quantity. 先圈出 find, show, prove, sketch, hence 或 explain。
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4. Keep exact values through intermediate steps; round once at the end.
5. Check sign, size, unit, domain and whether all solutions have been found.



例题一：从题意到完整解答 | Worked example 1

已知分解后求广义积分时应先检查什么？

检查分母零点是否落在积分区间；若有，普通定积分可能不存在。

Reading the mathematics. The first line identifies the structure; the middle lines show the transformation or calculation; the final line answers the question in context. Do not hide essential reasoning inside the calculator.

第 10 课（续） | P3 综合策略 | P3 synthesis

例题二：Cambridge 风格应用 | Cambridge-style application

How should exact answers be presented?

Keep π , e , logarithms and radicals exact unless a decimal accuracy is explicitly requested.

评分点如何产生 | Where marks come from

多数结构题把分数分给不同层次：建立正确模型或公式通常得到 method mark；正确代入并推进得到后续 method/accuracy marks；准确答案、单位、范围或结论得到 final accuracy

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Exam-ready summary. Define or identify $\rightarrow$ model $\rightarrow$ calculate with visible steps $\rightarrow$ interpret $\rightarrow$ check. Your final statement should answer the exact question asked, with appropriate accuracy and units.

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Detailed English explanation. Translate between words, symbols, diagrams and numerical values. State assumptions, keep exact values during working, and test the final result against the original domain, scale and context.

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第三个迁移任务 | Transfer task

How should exact answers be presented?

独立完成后对照：Keep π , e , logarithms and radicals exact unless a decimal accuracy is explicitly requested.

Exam check. Have you answered the command word, shown the decisive line of working, kept suitable accuracy, included units or direction, and considered every valid solution?